SANTHOSH V

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Dhanalakshmi Srinivasan Engineering College, Perambalur | B.E. Electronics & Communication Engineering (2023–2027) | CGPA: 7.69 | 0 Backlogs

CAREER OBJECTIVE

Electronics Engineering undergraduate (3rd Year) targeting internships in VLSI Design, Embedded Systems, or Core Electronics. Proficient in HDL coding (Verilog), circuit simulation (NI Multisim), and microcontroller programming (Arduino). Seeking to contribute to real-time hardware design, SoC architecture, and digital logic implementation in a hands-on engineering environment.

TECHNICAL SKILLS

HDL / Hardware: Verilog (Behavioral & Structural Modeling), NI Multisim (Analog/Digital Circuit Simulation)
Programming: C Programming, Python Programming, Verilog (Basics)
EDA & Prototyping: MATLAB, Arduino IDE, ESP8266,
Core Domains: VLSI Design, Digital Electronics, Analog & Mixed-Signal Circuits, Embedded Systems
OS & Tools: Linux (Ubuntu), Nmap(Basics)

EDUCATION

2023 – 2027 3rd Year | CGPA: 7.69 No Active Backlogs	**Dhanalakshmi Srinivasan Engineering College, Perambalur** B.E. – Electronics and Communication Engineering Anna University

PROJECTS

Air Gesture Mouse via IP Control (Linux)

- Problem: Physical touchpad dependency limits accessibility for remote PC control during presentations.
- Solution: Built a Python socket-based remote control system on Linux that streams motion data from a smartphone to control the host PC's cursor in real time via local IP.
- Tools: Python, Socket Programming, Linux (Ubuntu), IP Networking, Android Interface.
- Outcome: Delivered functional prototype with sub-100ms response; demonstrated at college-level Project Expo.

Smart Weather Monitoring System (Mini Project)

- Problem: Manual weather logging is error-prone and discontinuous in remote or rural environments.
- Solution: Interfaced DHT11 (temp/humidity) and BMP180 (pressure) sensors with Arduino; programmed data acquisition and live LCD display in C.
- Tools: Arduino Uno, DHT11, BMP180, I2C LCD, Arduino IDE (C).
- Outcome: Achieved continuous real-time multi-parameter display with sensor accuracy within ±1°C.

4-Bit Synchronous Up/Down Counter — Verilog HDL

- Problem: Design a configurable digital counter that operates in both increment and decrement modes using HDL.
- Solution: Implemented behavioral Verilog model for a synchronous 4-bit counter with active-high reset and direction control input; wrote a complete testbench.
- Tools: Verilog HDL, Xilinx ISim / EDA Playground (simulation & waveform analysis).
- Outcome: Validated all 16 state transitions and both operational modes through functional simulation; zero logic errors.

Smart Temperature Sensor — Arduino + LM35

- Programmed Arduino to read analog voltage from LM35 sensor, convert to Celsius via linear scaling, and display on 16x2 LCD with refresh every 500 ms.

- Tools: Arduino Uno, LM35, LCD (HD44780), C; Outcome: Stable readout with ±0.5°C accuracy over extended runtime.

WORKSHOPS & CERTIFICATIONS

- VLSI Chip Design Workshop — K. Ramakrishna Engineering College, Trichy (1 Day)
- Industrial Training — BSNL Telecom (1 Week)
- Digital VLSI Design — NPTEL
- MATLAB for Engineers — Self-Learned (Signal Processing)
- Multisim Circuit Simulation
- Introduction to IoT — NPTEL Certificate

ACTIVITIES & INTERESTS

- Active participant — College-Level Project Expo,Hackathons
- Tinkering with electronic gadgets and embedded modules
- Drawing & Sketching
- Reading Books

Declaration

I hereby declare that all information provided above is true and accurate to the best of my knowledge.